

LITERATURE ANSWER FOR THE DUAL ASQ VERSUS UASQ QUESTION

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STATUS

This is a literature-status packet, not a new proof. It records that the first open question in Garcia-Lirola–Rueda Zoca [1] is answered by the later paper of Abrahamsen–Hajek–Troyanski [2].

SOURCE QUESTION

In Section 6, Question 6.1 of [1], labelled **UASQASQequivalent** in the arXiv source, the authors ask:

Are UASQ and ASQ equivalent, at least, for dual Banach spaces?

The same paper proves that no dual Banach space is UASQ. Thus any dual ASQ Banach space gives a negative answer.

SUPPORTING ANSWER

The introduction of [2] explicitly discusses this question: after recalling UASQ, it says that Question 6.1 in the ASQ/UASQ literature asked whether dual ASQ spaces are UASQ, and that the construction of a dual ASQ space answers this in the negative. The decisive theorem is [2, Theorem 3.7]: a dual Banach space admits a dual ASQ norm if and only if it contains an isomorphic copy of c_0 .

Combining that theorem with the no-dual-UASQ theorem from [1] gives the answer to the source question:

dual ASQ $\not\Rightarrow$ UASQ.

Therefore UASQ and ASQ are not equivalent for dual Banach spaces.

SCOPE

This packet records only the first Section 6 question of [1]. It does not answer the later questions in the same source about uniformly discrete metric spaces M with $Lip(M)$ ASQ, or about isomorphic duality of $S(M, X)$.

REFERENCES

- [1] L. Garcia-Lirola and A. Rueda Zoca, *Unconditional almost squareness and applications to spaces of Lipschitz functions*, J. Math. Anal. Appl. 451 (2017), 117–131; arXiv:1605.04699.
- [2] T. A. Abrahamsen, P. Hajek, and S. Troyanski, *Almost square dual Banach spaces*, arXiv:1912.12519.